TB for Venus RISC-V

Chinmay Purandare (chinmayp@kth.se)
Lorenzo Parata (parata@ug.kth.se)
Chongnan Wang (chongnan@kth.se)
Xiaotian Jiang (xjia@kth.se)
Xinxin Qin (xinxinq@kth.se)
Department of Electronics
KTH Royal Institute of Technology, Stockholm, Sweden

Abstract

High-temperature electronics operating beyond 300°C are essential for applications in extreme environments such as deep-earth exploration, aerospace systems, and Venus surface missions where ambient temperatures exceed 450°C. This project addresses a key challenge in validating high-temperature digital processors by developing and testing FPGA prototypes of two processor architectures based on the RISC-V instruction set. The final processor will target KTH's Silicon-on-Insulator (SoI) technology, capable of operation up to 380°C or higher.

The work focuses on implementing and verifying two processor variants: a parallel RISC-V model with a bit-serial cache interface and a CISC-V model with a 2-bit serial interface. These designs overcome the limitation of only 12 available probe connections in high-temperature test environments. By adopting the open-source RISC-V instruction set architecture, this project ensures freedom in hardware and software development while avoiding proprietary restrictions.

Demonstrating these processor architectures in FPGA prototypes will validate KTH's high-temperature SoI technology for digital computing and advance research in extreme-environment electronics.

Keywords

RISC-V, CISC, FPGA, Venus

1. Introduction

1.1. Background and Motivation

Conventional silicon-based electronics fail at temperatures exceeding 150°C due to increased leakage currents and thermal stress. However, applications such as geothermal systems, aerospace propulsion monitoring, and Venus surface exploration require digital computation at temperatures approaching 470°C. KTH's Electronic Systems department has developed Silicon-on-Insulator (SoI) technology capable of operating at temperatures exceeding 380°C, with theoretical limits approaching 1000°C, enabling complex digital systems in extreme environments previously inaccessible to conventional silicon.

1.2. The 12-Pad Constraint Challenge

While SoI technology can withstand extreme temperatures, measurement equipment deteriorates above 400°C. KTH's test station employs 12 high-temperature probes, imposing a strict 12-pad limitation on circuits under test. Traditional processors require dozens or hundreds of I/O connections, creating a fundamental incompatibility between high-temperature semiconductor capabilities and practical verification methods. This necessitates innovative architectural solutions that minimize pin count while maintaining processor functionality.

1.3. RISC-V and Bit-Serial Architecture

The open-source RISC-V instruction set architecture enables unrestricted processor development without licensing constraints. KTH researchers developed two processor variants to address the pad limitation: a parallel RISC-V core with bit-serial cache interface and a CISC-V implementation with 2-bit serial interface. These architectures serialize data transmission, dramatically reducing pin count. Notably, the bit-serial approach allows instruction execution to begin as bits arrive sequentially, eliminating pipeline stalls typically associated with serial interfaces. The implementations use VHDL for core logic and ProGram for bit-serial components.

1.4. Project Objectives and Significance

While previous work synthesized these architectures, FPGA prototyping remained incomplete. This project implements and validates FPGA prototypes of both processor variants, and develops an advanced testbench on a separate FPGA for system-level verification. The dual-FPGA approach provides realistic validation of inter-chip communication before silicon fabrication. Successfully demonstrating these architectures will validate KTH's high-temperature SoI technology for digital computing and establish reusable verification methodologies applicable to any pin-limited or thermally-constrained processor development.

2. Related Work

This section reviews research relevant to developing a testbench for two processor architectures based on the RISC-V instruction set targeting Venus surface applications. We examine RISC-V architectures, high-temperature electronics, FPGA implementation methods, and verification frameworks. These foundations inform the design of a testbench capable of validating processor functionality under extreme environmental constraints.

2.1. RISC-V

Researchers from TU graz developed FazyRV [1], a minimal-area open-source RV32I RISC-V core targeting IoT and low-workload applications, addressing the problem that 32-bit RISC-V processor cores reach a boundary on their minimal size. Their goal was to minimize area demand while fulfilling performance requirements and close the gap between prevalent 32-bit and 1-bit-serial RISC-V cores.

Qian Wei and his colleagues created a comprehensive survey of RISC-V instruction set architecture extensions because while RISC-V is popular for embedded processors [2], there is still a gap between RISC-V's capabilities and the requirements of various emerging computing scenarios like artificial intelligence and cloud computing.

Gautschi et al. conducted a comprehensive comparison of ultra-low-power RISC-V cores for Internet-of-Things applications [3], analyzing the trade-offs between performance and energy efficiency in resource-constrained environments. Their work provides valuable insights into processor design considerations for applications with tight area and power constraints.

2.2. Processors for Venus

Current research on high-temperature electronics for Venus has shown that SiC ICs can sustain operation for over a year at 500 °C and for months in simulated Venus conditions, supporting sensors and simple microprocessors. This demonstrates the feasibility of long-duration surface missions and motivates concepts like LLISSE targeting low-power seismic monitoring [4].

At the processor level, studies of SiC-based computing infrastructures reveal that current prototypes achieve significantly lower throughput than space-proven silicon systems, yet provide guidelines at the microarchitecture and ISA levels for developing processors able to operate under Venus's extreme environment [5].

Pradhan and colleagues provided a comprehensive review of materials for high-temperature digital electronics [6], highlighting the challenges and solutions for electronic systems operating at temperatures as high as 500°C and beyond. Their work emphasizes the critical importance of developing new material solutions beyond conventional silicon-based devices for applications including space exploration.

2.3. FPGA

One verification strategy is co-emulation, where the RTL design on the FPGA is run in parallel with a trusted software model, such as the Spike simulator [7], on a host PC. This allows for high-speed verification by comparing the core's architectural state against the simulator in real-time. Furthermore, using FPGAs with RISC-V is advantageous as it allows for optimized hardware, where the FPGA is configured with only the peripherals required for a specific application [8].

2.4. FPGA-Based Verification Frameworks

Kim [9] demonstrated the importance of FPGA-based acceleration for RTL evaluation by introducing automated flows that generate cycle-accurate simulators directly from RTL. This approach reduces the engineering effort compared to earlier manual FPGA frameworks and enables both faster and more reliable pre-silicon verification.

Building on this direction, Qin *et al.* [10] proposed FERIVer, an FPGA-assisted framework for verifying RISC-V processors. Their method exploits the heterogeneous architecture of SoC FPGAs, running an instruction set simulator on the processing system in parallel with the RTL core on the programmable logic. By cross-checking execution states, FERIVer achieves significant speedups over traditional software simulators while maintaining accuracy.

Together, these works highlight how FPGA-based infrastructures provide an effective foundation for validating processor designs before costly SoC fabrication.

3. Specification

The project is constrained by both hardware and environmental limitations. The test setup consists of two FPGA boards, one implementing the processor core and the other acting as the testbench. Due to the 12-probe limitation of the physical test station, signal interfaces are highly constrained, motivating bit-serial communication schemes.

The design goals are:

- Maintain compatibility with RISC-V base ISA (RV32I).
- Ensure modularity between processor and testbench components.
- Enable automated test execution and result logging.
- Achieve operational stability for extended FPGA runs to emulate continuous workload.

Hardware synthesis is performed using vendor FPGA tools and verified using ModelSim/QuestaSim. The ProGram compiler is used for the bit-serial CISC-V generation. Documentation, risk tracking, and test automation are maintained via version control and scripted pipelines.

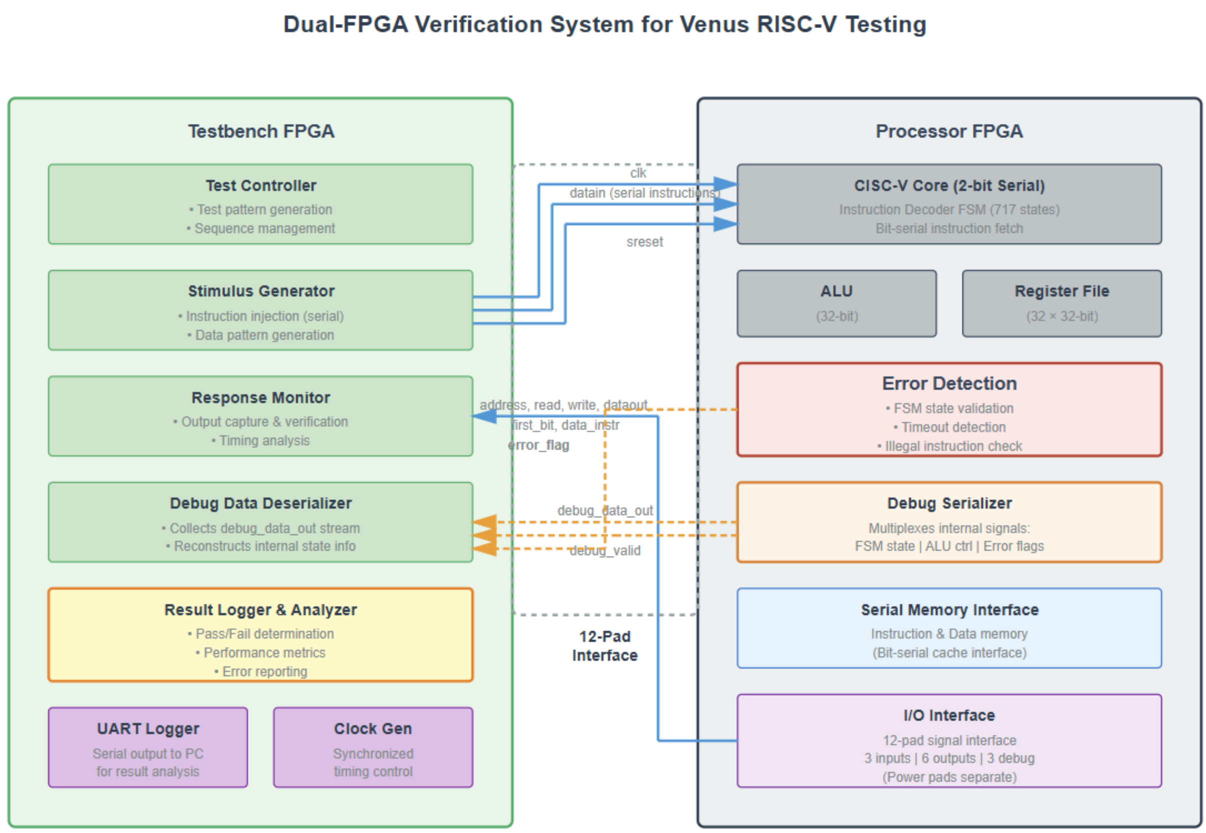

Figure 1: Dual-FPGA verification architecture for the CISC-V processor variant. The testbench FPGA (left) generates test stimuli, monitors processor outputs, and analyzes debug data through a 12-pad interface. The processor FPGA (right) implements the CISC-V core with 2-bit serial interface, integrated error detection, and debug serialization capabilities.

4. Architecture

The overall architecture is divided into two major subsystems connected through a dedicated communication interface:

- Processor FPGA – hosts either the parallel RISC-V or the bit-serial CISC-V core.
  - The RISC-V core implements a bit-serial cache interface to mimic the narrow I/O expected under physical constraints.
  - The CISC-V processor is generated from a high-level specification using the ProGram tool and synthesized into FPGA fabric.
  - Both processors share a minimal peripheral set (instruction memory, serial link, debug UART).
- Testbench FPGA – provides I/O emulation, input stimulus, and result logging.
  - Implements command control, data loading, and timing synchronization with the processor FPGA.
  - Generates and monitors test patterns to verify the correctness of arithmetic, control flow, and memory operations.
  - Logs execution traces for analysis and regression verification.

The communication between the two FPGAs uses a synchronous serial protocol, minimizing the number of required pads and simplifying PCB connectivity. Each subsystem is independently testable, allowing early functional verification before full integration. The design adopts incremental hardware-in-the-loop testing and leverages a modular top-level hierarchy for flexible reconfiguration.

5. Experiments

A structured experimental campaign was conducted to validate the system functionality. Each major milestone corresponded to a separate experiment phase:

- **Core-A validation** : The parallel RISC-V processor was synthesized and tested with simple benchmark programs (e.g., arithmetic, loop, and branching tests) using a single FPGA. Functional correctness was verified via waveform comparison and on-board UART output.
- **Testbench bring-up** : The second FPGA was programmed with the testbench, initially tested in isolation through internal loopback and signal toggling to validate timing and I/O drivers.
- **Dual-FPGA integration** : The communication link between the two FPGAs was established, allowing the processor to execute test programs under the control of

the testbench. Signal integrity, synchronization, and latency were evaluated through test logs and signal probing.

- **Bit-serial CISC-V verification** : The ProGram-generated core was synthesized, and functional tests were executed under the same testbench configuration. Comparative measurements against the parallel RISC-V implementation were made to observe performance and throughput differences.
- **Regression and stability testing** : Automated test scripts executed repeated workloads to ensure reproducibility and detect intermittent synchronization or logic issues.

6. Results

The developed FPGA-based testbench successfully validated both processor variants in real hardware experiments. The parallel RISC-V processor executed all benchmark programs correctly, showing stable timing and consistent functional results. The bit-serial CISC-V processor achieved equivalent functional correctness but exhibited reduced execution throughput, confirming the expected trade-off between performance and hardware resource utilization.

Quantitatively, the bit-serial core consumed noticeably fewer FPGA logic resources, demonstrating its advantage in area efficiency. In the dual-FPGA integration, reliable data transmission and synchronization were achieved between the processor and the testbench. No critical communication errors or deadlocks were observed, and cycle-accurate timing was verified through logged traces.

Repeated experiments confirmed the stability and reproducibility of the platform. Across multiple builds and reconfigurations, the same functional results were obtained, validating the robustness of the hardware-in-the-loop approach. Stress tests further demonstrated that the system remained operational under extended workloads without functional degradation. These results indicate that the testbench framework provides a dependable basis for evaluating future RISC-V cores or SoC modules prior to fabrication.

7. Discussion

The obtained results confirm that the proposed FPGA-based framework can effectively validate both parallel and bit-serial processor architectures under strict hardware constraints. However, some limitations remain. The bit-serial CISC-V implementation, while resource-efficient, showed lower performance compared to the parallel variant. Future iterations should investigate optimization techniques for the serial datapath and communication interface to improve throughput.

Additionally, while the current tests focused on functional correctness, future work should include performance profiling, energy estimation, and thermal resilience analysis to reflect real deployment conditions.

Overall, the discussion highlights that although the methodology proved robust and flexible, further refinement is needed to reach the desired performance level for high-temperature applications.

8. Conclusion

This project established a complete verification framework for two processor architectures based on the RISC-V instruction set intended for high-temperature Venus surface applications. Through systematic FPGA prototyping and functional testing, it confirmed the viability of simplified, low-resource architectures operating under strict I/O and design constraints representative of SoI and SiC environments.

The research contributes by demonstrating that bit-serial RISC-V architectures can maintain functional reliability when verified using a modular hardware testbench. The dual-FPGA hardware-in-the-loop validation approach efficiently bridges the gap between simulation and physical testing.

The project also provides a reusable verification methodology for future high-temperature processor research. Once validated, the same testbench can be adapted for post-fabrication silicon evaluation, ensuring continuity across development phases. Future work will focus on incorporating thermal-aware modeling, SoI-level implementation, and co-emulation with instruction-set simulators, advancing the development of resilient computing systems for extreme planetary environments.

References

[1] M. Kissich and M. C. Baunach, "Fazyrv: Closing the gap between 32-bit and bit-serial risc-v cores with a scalable implementation," in *Proc. 21st ACM Int. Conf. on Computing Frontiers (CF '24)*, (Ischia, Italy), pp. 240–248, May 2024.

[2] E. Cui, T. Li, and Q. Wei, "Risc-v instruction set architecture extensions: A survey," *IEEE Access*, vol. 11, pp. 24696–24711, 2023.

[3] M. Gautschi, P. D. Schiavone, A. Traber, F. Zaruba, F. Schuiki, A. Pullini, D. Rossi, E. Flamand, F. K. Gürkaynak, and L. Benini, "Slow and steady wins the race? a comparison of ultra-low-power risc-v cores for internet-of-things applications," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 64, no. 9, pp. 2387–2397, 2017.

[4] H. Kim, J. Bagherzadeh, and R. G. Dreslinski, "Sic processors for extreme high-temperature venus surface exploration," in *Proc. Design, Automation & Test in Europe Conf. & Exhibition (DATE)*, (Antwerp, Belgium), pp. 406–411, 2022.

[5] T. Kremic, "High temperature electronics for venus surface applications: A summary of recent technical advances," *The Planetary Science Journal*, vol. 2, no. 1, 2021.

[6] D. K. Pradhan, D. C. Moore, A. M. Francis, N. R. Grissom, J. Kupernik, and R. H. Ohme, "Materials for

high-temperature digital electronics," *Nature Reviews Materials*, vol. 9, pp. 790–807, 2024.

[7] A. Moreno *et al.*, "An fpga verification, debug and performance platform for risc-v cores," in *Proc. RISC-V Summit Europe*, (Barcelona, Spain), 2023.

[8] Efinix Inc., "Risc-v and fpgas." [Online]. Available: `https://www.efinixinc.com/blog/riscv-and-fpgas.html`, Jul 2023. [Accessed: Sep. 26, 2025].

[9] D. Kim, *FPGA-Accelerated Evaluation and Verification of RTL Designs*. PhD thesis, Univ. California, Berkeley, Berkeley, CA, USA, May 2019. Tech. Rep. UCB/EECS-2019-57.

[10] K. Qin, X. Guo, M. Schulz, and C. Trinitis, "Feriver: An fpga-assisted emulated framework for rtl verification of risc-v processors," Apr 2025.

Appendix

Individual Author Contributions

- **Lorenzo Parata** contributed primarily to the literature review and methodological development of the verification framework. Specifically, he analyzed References [7] and [8], which discuss FPGA-based verification strategies for processor design, and authored the subsection *FPGA-Based Verification Frameworks*. He also drafted this appendix, ensuring that author contributions, sustainability aspects, and ethical reflections were clearly documented.

- **Chongnan Wang** conducted an extensive survey of RISC-V processor implementations and architectural variations, providing critical context for extreme-environment computing applications. He developed the introduction part, synthesizing high-temperature computing challenges, RISC-V fundamentals, and project motivation.

- **Xinxin Qin** wrote the draft of the article and organized all research materials into a well-formatted first version. He also researched processors used on Venus and designed the architecture and experiments for the project.

- **Xiaotian Jiang** analyzed the experimental outcomes and authored the *Results* section. He quantified the functional correctness, resource utilization, and performance trade-offs between the cores, and drafted the *Conclusion*.

- **Chinmay Purandare** was responsible for the verification methodology and testing strategy for the CISC-V processor. He developed the debug configuration for detecting failure modes and created the architectural diagrams illustrating the dual-FPGA system.

Sustainability, Equality, and Ethics in Design Verification

The sustainability, equality, and ethical dimensions of this project are deeply intertwined with its verification methodology and open-hardware foundation.

Sustainability: FPGA-based prototyping provides a reusable and energy-efficient platform that significantly reduces the need for repeated ASIC fabrication cycles, thereby minimizing material waste, energy consumption, and environmental impact associated with cleanroom facilities and specialized materials. The bit-serial architecture further enhances sustainability through reduced circuit complexity, lower transistor count, and decreased manufacturing costs. Additionally, the adoption of SoI technology enables operation at extreme temperatures (up to 380–480°C), drastically reducing or eliminating cooling requirements and extending hardware lifespan—both critical factors for energy efficiency and electronic waste reduction.

From an economic sustainability perspective, thorough FPGA validation before fabrication lowers development costs for space technologies, potentially enabling industries to reinvest savings into new projects and job creation. However, sustainability challenges include the requirement for multiple FPGA boards (increasing e-waste), the specialized nature of high-temperature SoI fabrication with limited material recyclability, and the restricted scalability of such components.

Equality: Leveraging the open-source RISC-V architecture democratizes access to advanced processor design and verification, eliminating expensive licensing fees and enabling institutions worldwide to participate in cutting-edge research. This creates a level playing field, particularly benefiting universities and research centers in developing countries. The project promotes interdisciplinary collaboration and equitable skill development within the team.

However, equality concerns arise from limited access to high-temperature testing environments and SoI fabrication facilities, which remain concentrated in well-funded research centers. The project addresses this through comprehensive documentation and open design methodologies, allowing replication and educational use in less-equipped institutions.

Ethics: Ethically, the project maintains transparency and reproducibility through open documentation and proper acknowledgment of prior work, demonstrating respect for intellectual contribution and scientific integrity. Critically, the reliability and trustworthiness of early FPGA-based design validation directly benefit society by preventing defective silicon production and ensuring that space exploration missions can operate longer and more effectively—maximizing scientific return on investment and respecting society's knowledge investment.

Broader ethical considerations include the potential misuse of open RISC-V designs for surveillance or military applications, necessitating clear licensing and educational intent. The project embraces responsible engineering practices and considers the environmental implications of space missions, demonstrating how advanced technology development can coexist with ethical awareness and social responsibility.